

KRZYSZTOF KOTYSZ

SOFTWARE & INFRASTRUCTURE ENGINEER | PYTHON | LINUX | AUTOMATION

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PROFESSIONAL SUMMARY

Software and infrastructure engineer working across Python, Linux, distributed observatories, and scientific instrumentation. For BHTOM, coordinated observations across more than 100 telescopes and helped standardise and containerise automatic CCD reduction; during an on-site visit to Las Cumbres Observatory, integrated the complete processing pipeline with Prefect. Independently built the current ALPS public website and develops software for telescope operation and automation. Brings 8+ years of experience from systems where software must work reliably with remote hardware and scientifically validated data.

TECHNICAL SKILLS

Workflow engineering: integrated the full BHTOM pipeline with Prefect; worked on unifying and containerising automatic CCD reduction with Docker.

Distributed operations: BHTOM portal and more than 100 telescopes worldwide; Linux services, remote systems, networking, NFS, VPN, and operational diagnosis.

Software development: Python, JavaScript, and Fortran; scientific analysis tools, observatory-support software, web applications, databases, and automation.

Instrument integration: software for a 60-cm Cassegrain telescope; automation and robotisation of a 32-cm telescope; CCD cameras, sensors, and field troubleshooting.

Technical communication: university teaching in Python, Fortran, Linux, mathematics, statistics, and astronomy; documentation and international collaboration.

EXPERIENCE

BHTOM Engineering Specialist and Contractor

Feb 2024 - Present

University of Warsaw - Astronomical Observatory

employment and project contracts through Dec 2026

- Operated BHTOM, a portal used to synchronise, supervise, and conduct observations across multiple observatories; was responsible for monitoring a worldwide network of more than 100 telescopes.
- Worked on unifying the automated CCD image-reduction system and packaging it in containers for distributed observatory deployments.
- During an on-site collaboration at Las Cumbres Observatory in Goleta, California (8-17 Feb 2025), integrated the complete existing BHTOM pipeline with Prefect and resolved integration details directly with the LCO team.

Doctoral Researcher and Research Assistant

2018 - Present

University of Wrocław - Institute of Astronomy

- Authored software supporting observations with the 60-cm Cassegrain telescope and contributed to modernising the Białków Observatory; also works on automation and robotisation of its 32-cm telescope.
- Built analysis programs for TESS sectors 1-13, screened more than 4,000 light curves, and selected 90 objects for detailed Fourier analysis of pulsation modes and amplitude or period changes.
- Operated and diagnosed professional instruments during 168 observing nights: 87 at Białków and 81 across four campaigns at remote observatories in Chile.
- Prepared a spectroscopic dataset for more than 70 targets to complement photometric work on the difficult classification of HADS and SX Phoenicis stars.
- Secured and delivered NCN PRELUDIUM-17 project 2019/33/N/ST9/01818, covering proposal preparation, data acquisition, analysis, documentation, and reporting.

PROJECTS

Distributed Observation and Workflow System - BHTOM 2024 - Present

- Supported a portal that synchronises observations performed by many observatories and supervised the operational state of more than 100 telescopes distributed worldwide.
- Worked on a uniform, containerised automatic CCD-reduction workflow; integrated the complete pipeline with Prefect during direct engineering work with LCO in California.

ALPS - Public Web and Measurement Infrastructure 2018 - Present

- Independently designed and implemented the complete current ALPS public website: station map, live network status, SQM light-pollution measurements, environmental data, and project content.
- Contributes across the wider multi-site system connecting measurement stations in Poland, sensors, software, infrastructure, and data processing.

Observatory Software and Telescope Automation

- Writes custom tools that support observers using a 60-cm Cassegrain telescope and incorporates feedback gathered during 87 nights of operation at Białków.
- Contributes to automation and robotisation of a 32-cm telescope, joining control software, instrument testing, fault diagnosis, and data-quality checks.

TEACHING EXPERIENCE

University Teaching - University of Wrocław

- Taught university classes in mathematics, astronomy, and statistics, with exercises built around quantitative analysis and scientific problem-solving.
- Led practical classes in Python, Fortran, and Linux, teaching students to implement numerical tasks and work in scientific computing environments.
- Guided students through analytical and programming exercises and translated specialist concepts into clear technical explanations.

EDUCATION

PhD in Astronomy (in progress) 2018 - Present
University of Wrocław large-scale time-series and spectroscopic data analysis

MSc in Astronomy 2016 - 2018
University of Wrocław satellite data correction and stellar variability analysis

BSc in Astronomy 2013 - 2016
University of Wrocław precision astrometry using more than a decade of observations

OPEN-SOURCE PROJECTS AND PUBLICATIONS

- Public GitHub portfolio includes *lcview* and *astroplanner*, providing reviewable code samples from independent scientific-software projects.
- Co-author of 13 peer-reviewed papers in journals including MNRAS, Astronomy & Astrophysics, The Astrophysical Journal, and Acta Astronomica, plus two proceedings.

LANGUAGES

Polish: native | English: professional working proficiency; international technical collaboration and scientific writing
Spanish: basic; semester course and simple travel situations